Capstone Project - Progress Report - 2

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Location: Hyderabad, Telangana, India
Course: EMTech - Cloud Computing
Subject Code: EHS 5104
Project Type: Capstone project with CI/CD with Github Actions

Change log

- **v1.1.0 (Dec 15, 2025)**
  - Update C++11 compiler to C++17 in conan package manager
  - Added support for various libraries like
    - openssl-v3.5.0
    - websocketpp-v0.8.2
    - boost-v1.83.0
    - rabbitmq-c-v0.15.0
    - grpc-v1.72.0
  - gitHub release tag v1.1.0

Overview

`ct-recon` is a C++ project designed for modular, containerized development and deployment. It integrates modern C++ practices with a streamlined CI/CD pipeline using GitHub Actions and Docker.
GitHub Actions is often chosen over Jenkins for modern CI/CD pipelines because it offers native GitHub integration, easier setup, and lower maintenance overhead—especially for cloud-native and open-source projects.

- The project is hosted on GitHub, so Actions integrates seamlessly.
- No need to manage Jenkins infrastructure.
- YAML workflows are version-controlled alongside the code.
- Docker and Conan steps are easily automated with reusable actions.
- Ideal for a capstone project with limited deployment complexity.

Github Repository

Click here for view github repository

Key Technologies

Category	Tools/Technologies
Language	C++ (C++11 and beyond)
Build System	CMake (cmake version 3.22.1)
Package Manager	Conan (Conan version 2.22.1)
CI/CD	GitHub Actions
Containerization	Docker
Scripting	Shell, Python

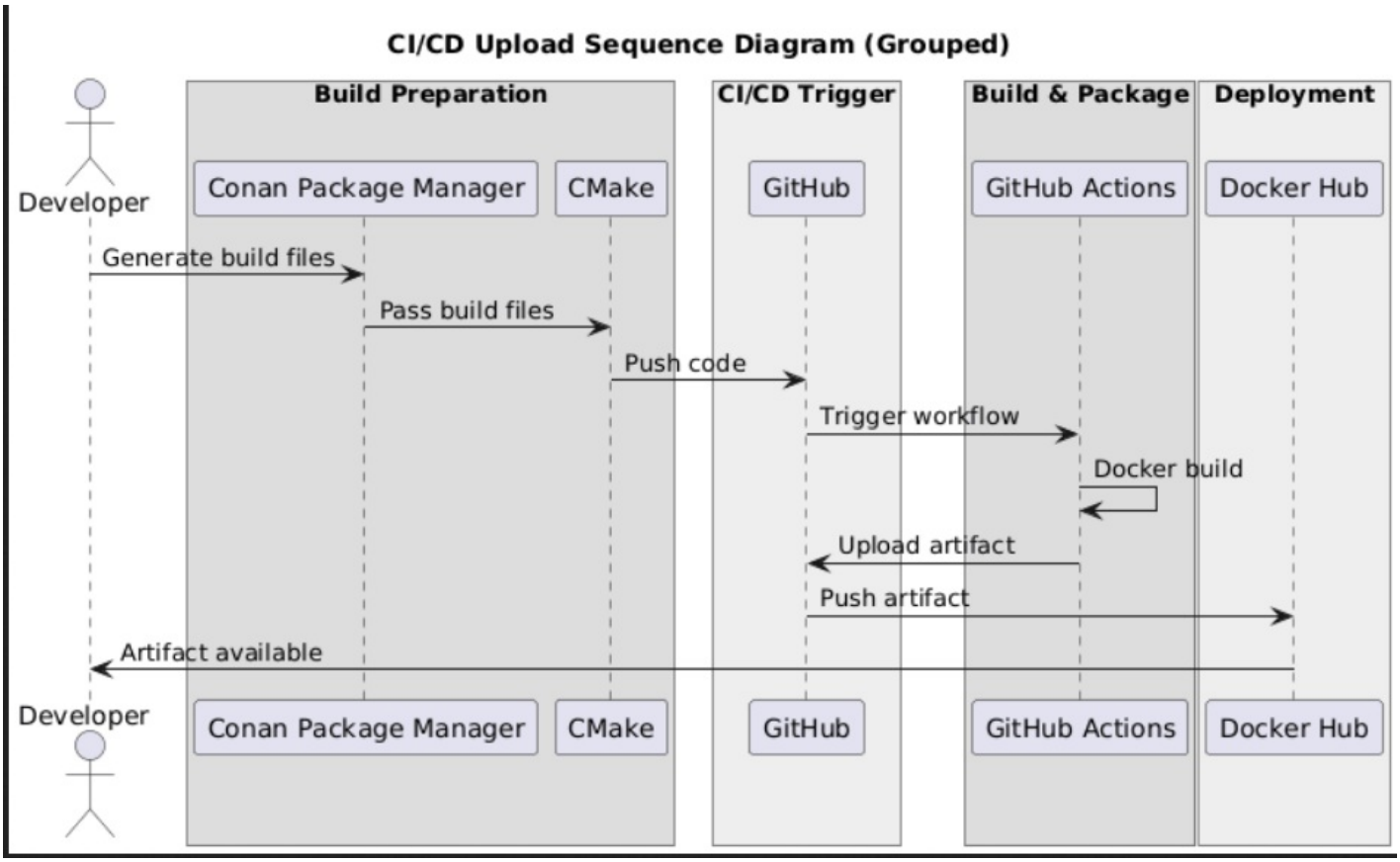
Project Structure

- `source/`, `include/`, `test/` – Core source and test files
- `.github/workflows/` – GitHub Actions CI/CD definitions
- `conanfile.py` – Conan dependency management
- `Dockerfile` – Docker image definition
- `run.sh` – Local test script

```
2025-11-01 21:21:32 usomgupta@usomgupta ~/gitHub/ct-reconmain $ tree -L 3
.
├── CMakeLists.txt
├── CMakeUserPresets.json
├── Dockerfile
├── MODULE.bazel
├── README.md
├── build
│   ├── Release
│   │   ├── CMakeCache.txt
│   │   ├── CMakeFiles
│   │   ├── Makefile
│   │   ├── cmake_install.cmake
│   │   ├── ctReconServiceToolBin
│   │   ├── generators
│   │   ├── libctReconServiceTool.so
│   │   └── metadata
│   └── build
│       └── Release
├── conanfile.py
├── include
│   ├── BUILD
│   └── testInc.h
├── input.txt
├── mtechDocumentsSlides
├── run.sh
├── source
│   ├── BUILD
│   ├── CMakeLists.txt
│   ├── MODULE.bazel
│   ├── main.cpp
│   └── testSrc.cpp
└── test
    ├── CMakeLists.txt
    └── main.cpp

11 directories, 22 files
```

CI/CD Workflow Diagram (PlantUML)




```
@startuml
title CI/CD Workflow for ct-recon

actor Developer

box "Build Preparation" #DDDDDD
participant "Conan"
participant "CMake"
end box

box "Source Control" #EEEEEE
participant "GitHub"
end box

box "Automation" #DDDDDD
participant "GitHub Actions"
end box

box "Deployment" #EEEEEE
participant "Docker Hub"
end box

Developer -> Conan : Install dependencies
Conan -> CMake : Generate build files
CMake -> GitHub : Push source code
GitHub -> GitHub Actions : Trigger CI workflow
GitHub Actions -> GitHub Actions : Build Docker image
GitHub Actions -> Docker Hub : Push image
Docker Hub -> Developer : Image available for use

@enduml
```

Docker Usage

```
# Build Docker image
docker build . -t ct-recon:<commit-hash-or-version>

# Login to Docker Hub
docker login -u usomgupta

# Run the container
docker run ct-recon:v1 ./ct-recon --input data.txt --verbose

# Shell into the container
docker run -it [image ID] bash

# Link to Docker Hub:
[Docker Hub: usomgupta/ct-recon](https://hub.docker.com/r/usomgupta/ct-recon)
```

CI/CD Workflow

The CI/CD pipeline is triggered on code push and automates the build and deployment process using GitHub Actions and Docker Hub.

- Uses: ubuntu-latest image
- Checkout from repository
- Login into Docker Hub
- Set up Docker Build
- Build and push Docker image
- Upload build artifacts

```yaml
name: CI for docker/build-push-action

on:
  push:
    branches:
      - main

permissions:
  contents: write
  issues: write
  pull-requests: write

jobs:
  build-and-push:
    runs-on: ubuntu-22.04  # Pin to a stable runner version
    permissions:
      contents: write
      pull-requests: read

    steps:
      - name: Checkout repository
        uses: actions/checkout@v4
        with:
          fetch-depth: 1  # Shallow clone for faster checkout

      - name: Log in to Docker Hub
        uses: docker/login-action@v3
        with:
          registry: ${{ env.REGISTRY }}
          username: ${{ secrets.DOCKER_USERNAME }}
          password: ${{ secrets.DOCKERHUB_TOKEN }}

      - name: Set up Docker Buildx
        uses: docker/setup-buildx-action@v3
        with:
          driver: docker-container
          driver-opts: |
            image=moby/buildkit:latest

      - name: Build and push Docker image
        uses: docker/build-push-action@v5
        with:
          builder: ${{ steps.buildx.outputs.name }}
          platforms: linux/amd64  # Use single platform unless multi-arch is needed
          context: .
          push: true
          tags: ${{ secrets.DOCKER_USERNAME }}/repo:v1
          cache-from: type=gha
          cache-to: type=gha,mode=max

      - name: Upload C++ build artifacts
        uses: actions/upload-artifact@v4
        with:
          name: cpp-build-artifacts
          path: |
            build/libmylib.so
            build/myprogram
```

Local Testing

To run the project locally:

```
sh run.sh
```

Testing the docker artifact

```
PS C:\Users\ritup> docker pull usomgupta/repo:v1
PS C:\Users\ritup> docker pull usomgupta/repo:v1
v1: Pulling from usomgupta/repo
d5c79deda2be: Pull complete
aa46819fa765: Pull complete
f0623a8556e7: Pull complete
8a422c469e9b: Pull complete
7d1b2d71253c: Pull complete
4f4fb700ef54: Pull complete
8e2f2dc427c8: Pull complete
888a6fd2a6da: Pull complete
236f3b202ef5: Pull complete
Digest: sha256:fa3d686e4c1391ab6db754bbadb76d5823d690d004933d7223846496d94b0074
Status: Downloaded newer image for usomgupta/repo:v1
docker.io/usomgupta/repo:v1
PS C:\Users\ritup> docker run -it  bash^C
PS C:\Users\ritup> docker images
REPOSITORY       TAG       IMAGE ID       CREATED        SIZE
usomgupta/repo   v1        fa3d686e4c13   55 years ago   423MB
PS C:\Users\ritup> docker run -it fa3d686e4c13 bash
root@d708f1523a67:/app# ls
CMakeLists.txt         Dockerfile    README.md  conanfile.py  input.txt            run.sh  test
CMakeUserPresets.json  MODULE.bazel  build      include       mtechDocumentsSlides  source
root@d708f1523a67:/app# cd build/
Release/ build/
root@d708f1523a67:/app# cd build/Release/
root@d708f1523a67:/app/build/Release# ls
CMakeCache.txt  CMakeFiles  Makefile  cmake_install.cmake  ctReconServiceToolBin  generators  libctReconServiceTool.so  metadata
root@d708f1523a67:/app/build/Release# ./ctReconServiceToolBin
foo is called from main
Execution begin for functionreverseNumber
The reversed negative number 321
Execution begin for functionreverseNumber
The reversed negative number -321
root@d708f1523a67:/app/build/Release# cd ../../
root@d708f1523a67:/app# ls
CMakeLists.txt         Dockerfile    README.md  conanfile.py  input.txt            run.sh  test
CMakeUserPresets.json  MODULE.bazel  build      include       mtechDocumentsSlides  source
```

Summary

The CI/CD pipeline is triggered on every code push. It automates build, test, and deployment using GitHub Actions, and publishes Docker images to Docker Hub for seamless distribution.
`Future work` : **I will try to deploy artifacts in jfrog**